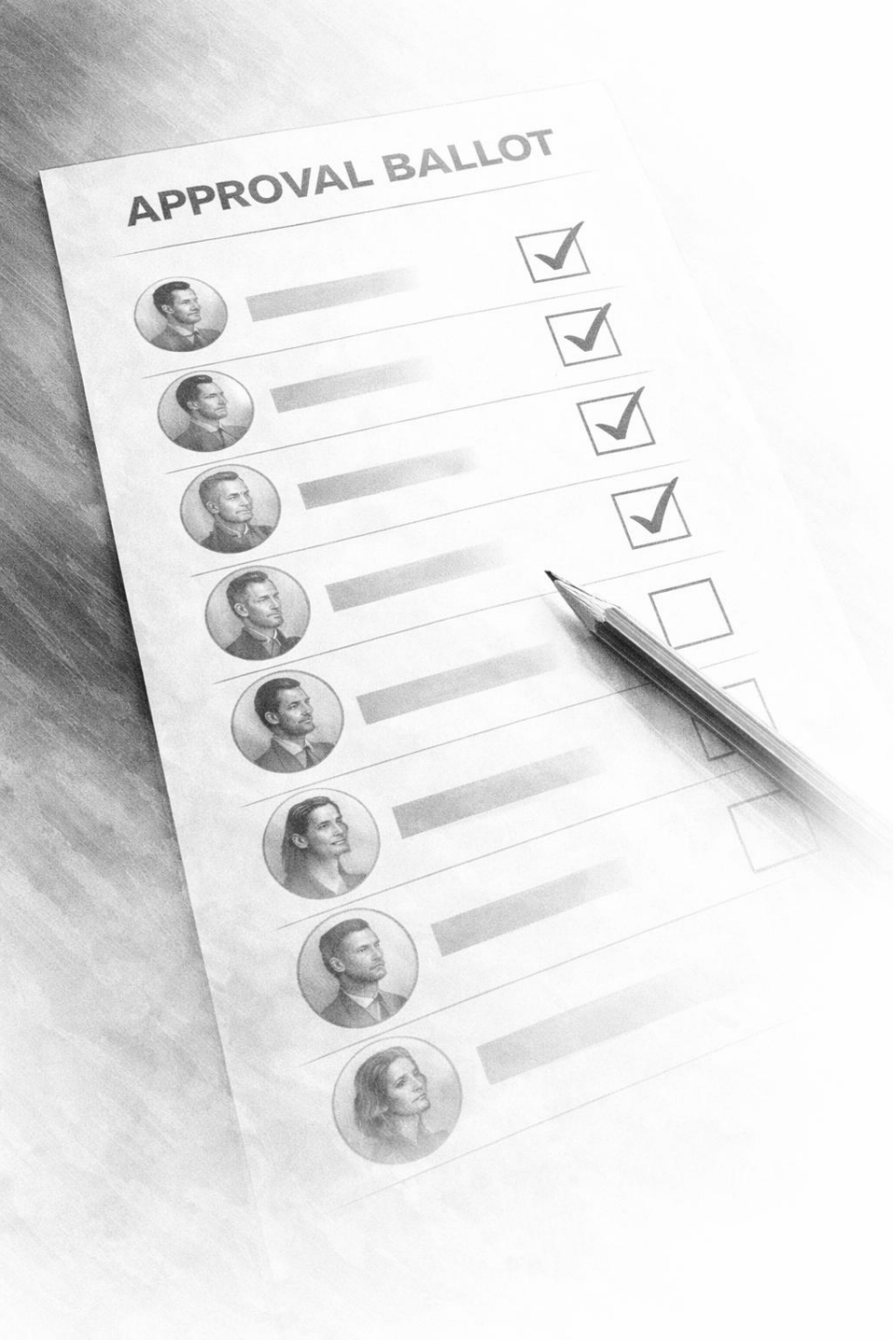

Features of Approval Elections

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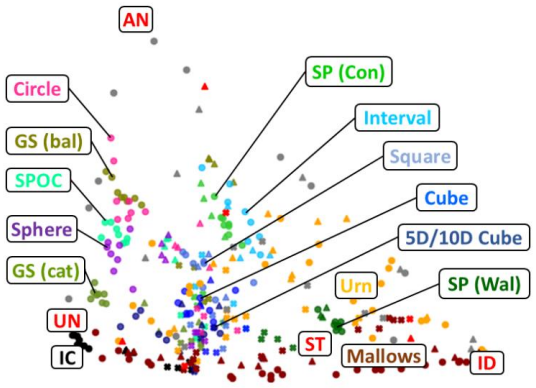
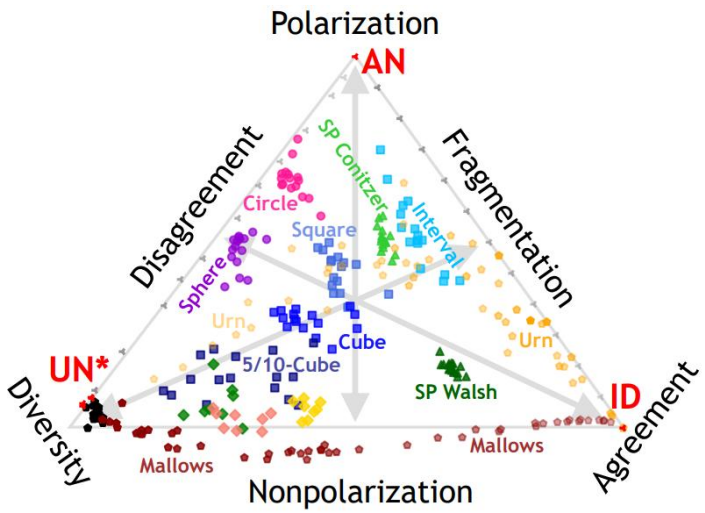
Krzysztof Sornat
Stanisław Szufa
Tomasz Wąs

Election Indices

Index: A function that associates an election with a value in [0,1]





Possible to make maps using feature distances

Measures an extent to which an election has some feature

Agreement: Are voters' views similar?
Diversity: Do the votes cover all possible views?
Polarization: Are there two groups with conflicting views that agree internally

„Independent" of election size, helps to compare elections with different number of candidates and voters

Normalization: This will only work if our indices are properly normalized

Global

Saturation-Based

Approval Indices: Normalization

Saturation: Average number of candidates approved by a single voter

Saturation affects how we interpret elections and their features

candidates

voters

$$\begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

Maximally diverse 5x5 election for saturation = 1/5

Not so diverse if any number of approvals is allowed

Index:

$$f(E) = g(E) \,/\, max_{E'}\, g(E')$$

Global normalization: An index is normalized over all possible elections of a given size

Saturation-Based Normalization: Normalization over elections with the same saturation

Intuition: Elections of the same nature, but different saturation have the same index value. Not a formal concept. We can only argue for it intuitively, or show it fails

Indices

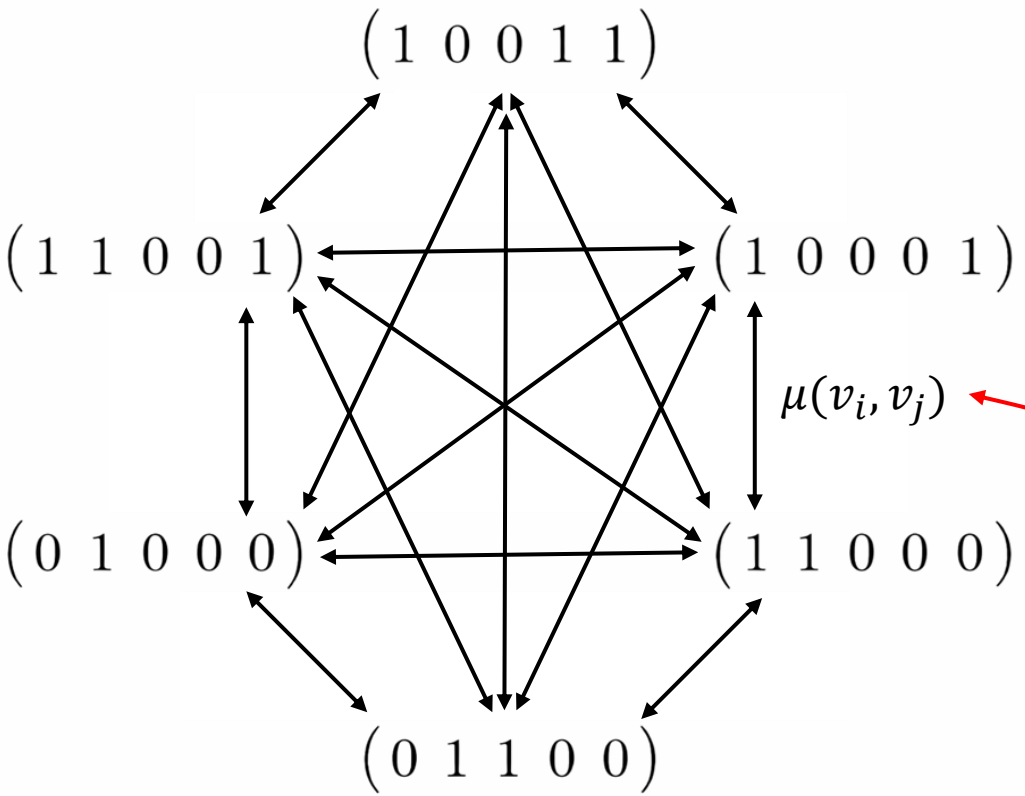
Agreement Indices: Pairwise Agreements

Global Agreement: All voters have similar views and generally agree with each other

Local Agreement: There are groups of voters that largely agree with each other (but not necessarily with others)

Saturation-Based Normalization: Elections of the same nature, but different saturation should have the same index value.

Test on Data: Check how a given index performs on example elections. Does it match intuition?



Definition 5.6. *For an election $E = (C, V)$, we define Hamming, Jaccard, PCC, and PCC$^+$ Pairwise Agreement Indices:*

$$\text{pair-agr}(E) = 1 - \frac{\sum_{u \in V} \sum_{v \in V} \text{ham}(u,v)}{2|V|^2|C| \cdot \text{satr}(E)(1-\text{satr}(E))},$$

$$\text{jacc-agr}(E) = \frac{1}{|V|^2} \sum_{u \in V} \sum_{v \in V} (1 - \text{jacc}(u,v)),$$

$$\text{pcc-agr}(E) = \frac{1}{|V|^2} \sum_{u \in V} \sum_{v \in V} \text{pcc}(u,v),$$

$$\text{pcc}^+\text{-agr}(E) = \frac{1}{|V|^2} \sum_{u \in V} \sum_{v \in V} \max(0, \text{pcc}(u,v)).$$

Diversity and Polarization

Votes are different from each other

Votes cover the space of possible preferences

Votes can be split into **two** groups

Internal agreement, disagreement between groups

partitioning into groups does not lead to great increase of agreement within groups

partitioning into two groups gives near-perfect agreement in these groups

clustering-based definitions

Diversity and Polarization

spectral clustering heuristic
to compute these values

Votes are different from each other

Votes cover the space of possible preferences

partitioning into groups does not lead to great increase of agreement within groups

Votes can be split into **two** groups

Internal agreement, disagreement between groups

partitioning into two groups gives near-perfect agreement in these groups

clustering-based definitions

Definition 6.1. *For an election $E = (C, V)$, and agreement index a, we define the a-*Diversity *index as:*

$$a\text{-div}(E) = 1 - \frac{1}{5}\sum_{k=1}^{5} \max_{\{V_1,\ldots,V_k\}\in\mathcal{P}_k(V)} \sum_{i=1}^{k} \frac{|V_i|}{|V|} a(C, V_i).$$

Definition 6.2. *For an election $E = (C, V)$, and agreement index a, we define the a-*Polarization *index as:*

$$a\text{-pol}(E) = \max_{\{V_1,V_2\}\in\mathcal{P}_2(V)} \sum_{i=1}^{2} \frac{|V_i|}{|V|} a(C, V_i) - a(E).$$

Clustering-Based Polarization

Definition 6.2. *For an election $E = (C, V)$, and agreement index a, we define the a-*Polarization *index as:*

$$a\text{-pol}(E) = \max_{\{V_1, V_2\} \in \mathcal{P}_2(V)} \sum_{i=1}^{2} \frac{|V_i|}{|V|} a(C, V_i) - a(E).$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

Low agreement

Clustering-Based Polarization

Definition 6.2. *For an election $E = (C, V)$, and agreement index a, we define the a-*Polarization *index as:*

$$a\text{-pol}(E) = \max_{\{V_1, V_2\} \in \mathcal{P}_2(V)} \sum_{i=1}^{2} \frac{|V_i|}{|V|} a(C, V_i) - a(E).$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

High agreement

High agreement

Clustering-Based Polarization

Definition 6.2. *For an election $E = (C, V)$, and agreement index a, we define the a-*Polarization *index as:*

$$a\text{-pol}(E) = \max_{\{V_1, V_2\} \in \mathcal{P}_2(V)} \sum_{i=1}^{2} \frac{|V_i|}{|V|} a(C, V_i) - a(E).$$

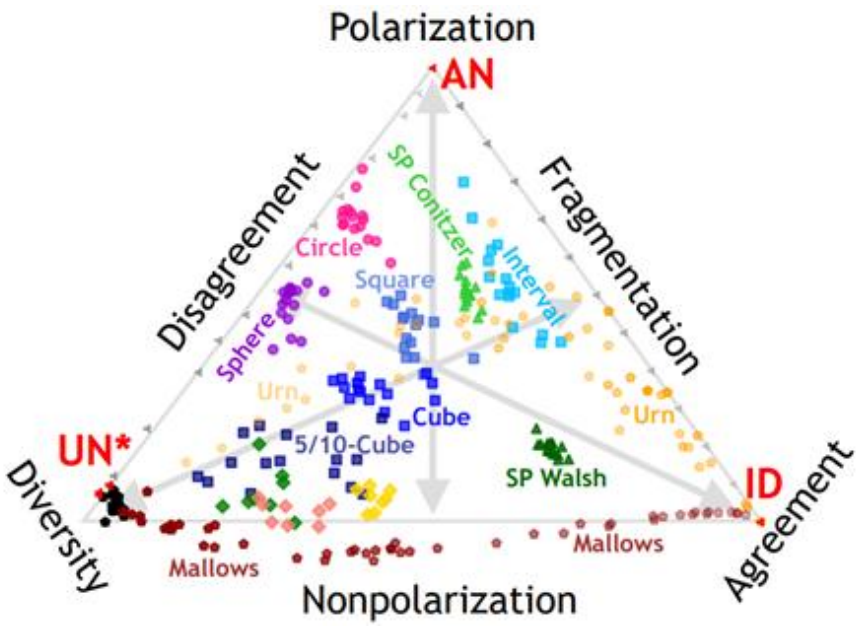
$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix} = \text{polar-ization}$$

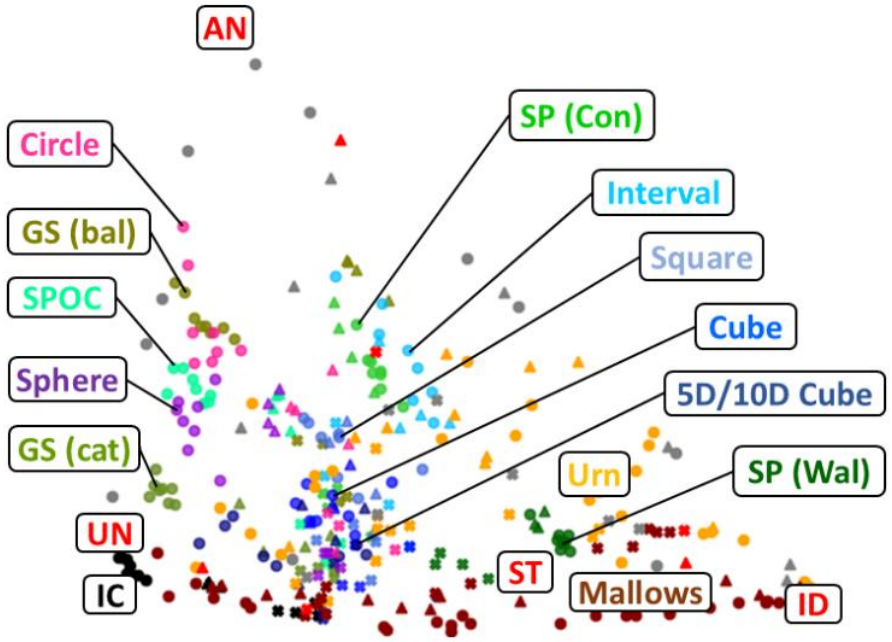
		av-agr	cntr-agr	pair-agr	pcc-agr	jacc-agr	pcc$^+$-agr	cntr-div	pcc-div	out-div	cntr-pol	pcc-pol	pair-pol
	resampling experiment:												
1.	$^1/_3$-ID	1.00 ±0.00	1.00 ±0.00	1.00 ±0.00	1.00 ±0.00	1.00 ±0.00	1.00 ±0.00	0.00 ±0.00	0.00 ±0.00	0.00 ±0.00	0.00 ±0.00	0.00 ±0.00	0.00 ±0.00
2.	2-Party	0.00 ±0.00	0.00 ±0.00	0.00 ±0.00	0.00 ±0.00	0.50 ±0.00	0.50 ±0.00	0.20 ±0.00	0.25 ±0.02	0.10 ±0.00	1.00 ±0.00	1.00 ±0.00	1.00 ±0.00
3.	N(2-Party, 0.6)	0.09 ±0.01	0.08 ±0.01	0.01 ±0.00	0.02 ±0.00	0.35 ±0.01	0.10 ±0.01	0.66 ±0.02	0.82 ±0.01	0.29 ±0.00	0.25 ±0.10	0.17 ±0.01	0.24 ±0.01
4.	3-Party	0.33 ±0.00	0.00 ±0.00	0.00 ±0.00	0.00 ±0.00	0.33 ±0.00	0.33 ±0.00	0.33 ±0.00	0.31 ±0.01	0.13 ±0.00	0.33 ±0.00	0.50 ±0.00	0.63 ±0.00
5.	4-Party	0.50 ±0.00	0.00 ±0.00	0.00 ±0.00	0.00 ±0.00	0.25 ±0.00	0.25 ±0.00	0.47 ±0.04	0.40 ±0.00	0.15 ±0.00	0.25 ±0.00	0.33 ±0.00	0.43 ±0.00
6.	($^1/_3$, $^1/_3$)-2-Party	0.33 ±0.00	0.24 ±0.00	0.10 ±0.00	0.11 ±0.00	0.56 ±0.00	0.56 ±0.00	0.15 ±0.00	0.18 ±0.00	0.09 ±0.00	0.76 ±0.00	0.89 ±0.00	0.99 ±0.00
7.	Cyclic	0.00 ±0.00	0.00 ±0.00	0.00 ±0.00	0.00 ±0.00	0.39 ±0.00	0.25 ±0.00	0.46 ±0.00	0.57 ±0.01	0.22 ±0.00	0.50 ±0.00	0.33 ±0.00	0.58 ±0.00
8.	Diagonal	0.97 ±0.00	0.00 ±0.00	0.00 ±0.00	0.00 ±0.00	0.02 ±0.00	0.02 ±0.00	0.97 ±0.00	0.97 ±0.00	0.63 ±0.01	0.02 ±0.00	0.02 ±0.00	0.01 ±0.00
9.	Triangle	0.50 ±0.00	0.49 ±0.00	0.33 ±0.00	0.50 ±0.00	0.51 ±0.00	0.50 ±0.00	0.41 ±0.00	0.32 ±0.00	0.18 ±0.00	0.01 ±0.00	0.14 ±0.00	0.47 ±0.00
10.	N(Tri,0.6)	0.21 ±0.01	0.20 ±0.02	0.06 ±0.00	0.15 ±0.02	0.33 ±0.01	0.17 ±0.02	0.89 ±0.01	0.77 ±0.03	0.26 ±0.00	0.00 ±0.02	0.05 ±0.00	0.39 ±0.01
11.	$^1/_2$-ID/IC	0.50 ±0.01	0.49 ±0.01	0.26 ±0.01	0.25 ±0.01	0.51 ±0.00	0.30 ±0.00	0.41 ±0.01	0.50 ±0.00	0.19 ±0.00	0.07 ±0.01	0.26 ±0.02	0.45 ±0.01
12.	$^1/_2$-IC	0.10 ±0.01	0.10 ±0.01	0.02 ±0.00	0.02 ±0.00	0.34 ±0.01	0.07 ±0.00	0.81 ±0.01	0.91 ±0.00	0.29 ±0.00	0.07 ±0.01	0.05 ±0.00	0.18 ±0.00
13.	$^1/_4$-IC	0.49 ±0.01	0.00 ±0.00	0.02 ±0.00	0.02 ±0.00	0.16 ±0.00	0.07 ±0.00	0.97 ±0.00	0.90 ±0.00	0.31 ±0.00	0.01 ±0.01	0.05 ±0.00	0.16 ±0.00
14.	Lin-IC	0.08 ±0.01	0.06 ±0.01	0.01 ±0.00	0.09 ±0.02	0.31 ±0.01	0.15 ±0.03	0.96 ±0.01	0.83 ±0.03	0.27 ±0.00	0.00 ±0.01	0.04 ±0.00	0.37 ±0.01

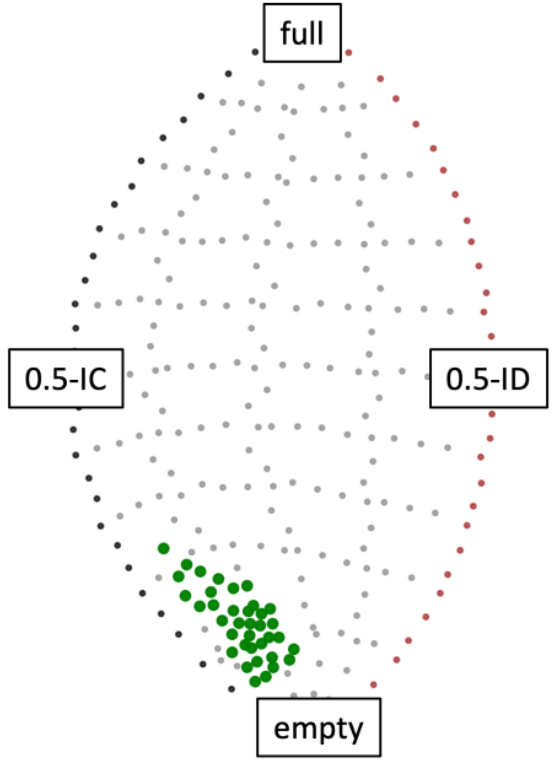
These three indices typically sum up to a value very close to 1, hence forming an approximate simplex

Map

Maps of Elections: Ordinal versus Approval



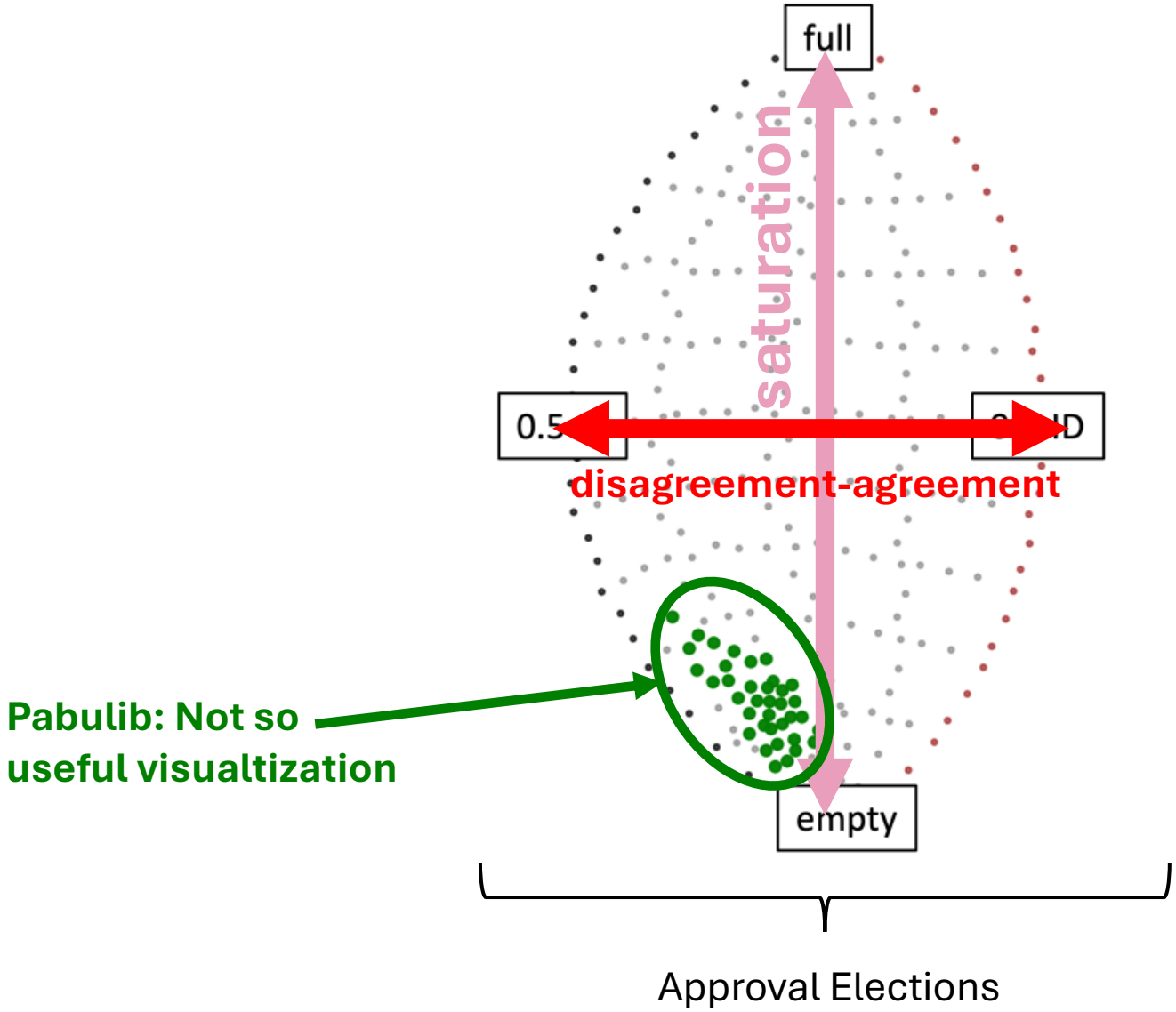




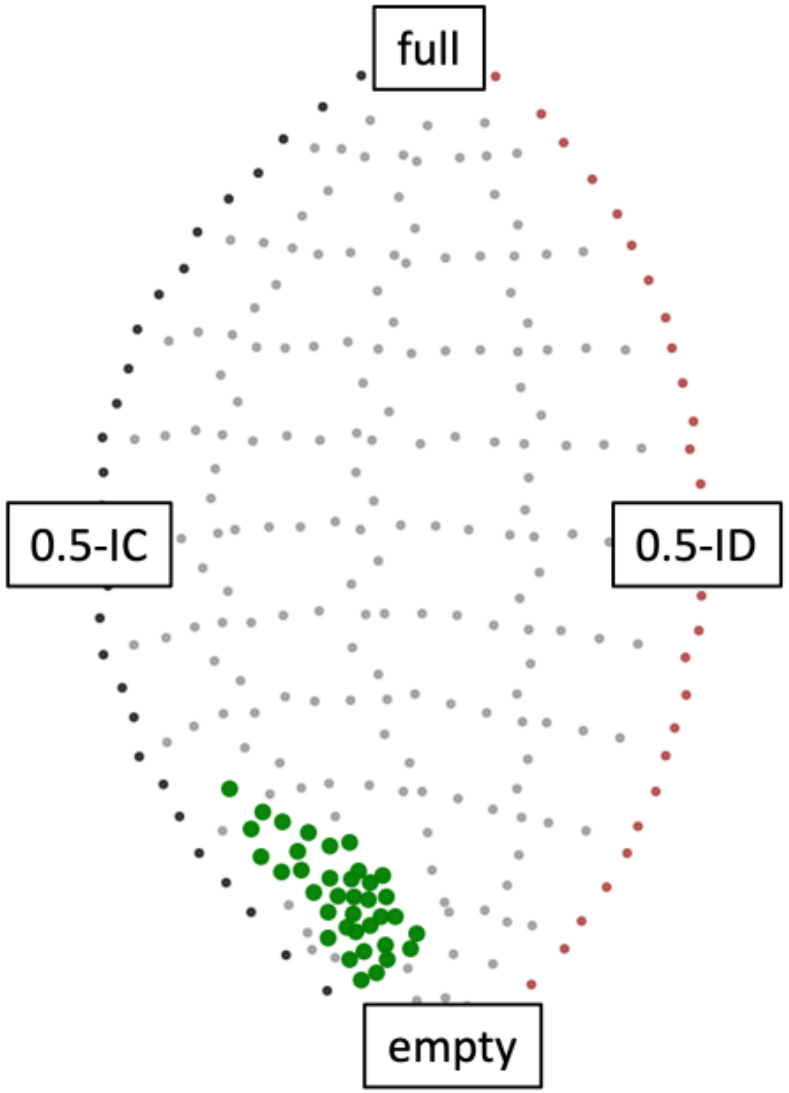
Ordinal Elections

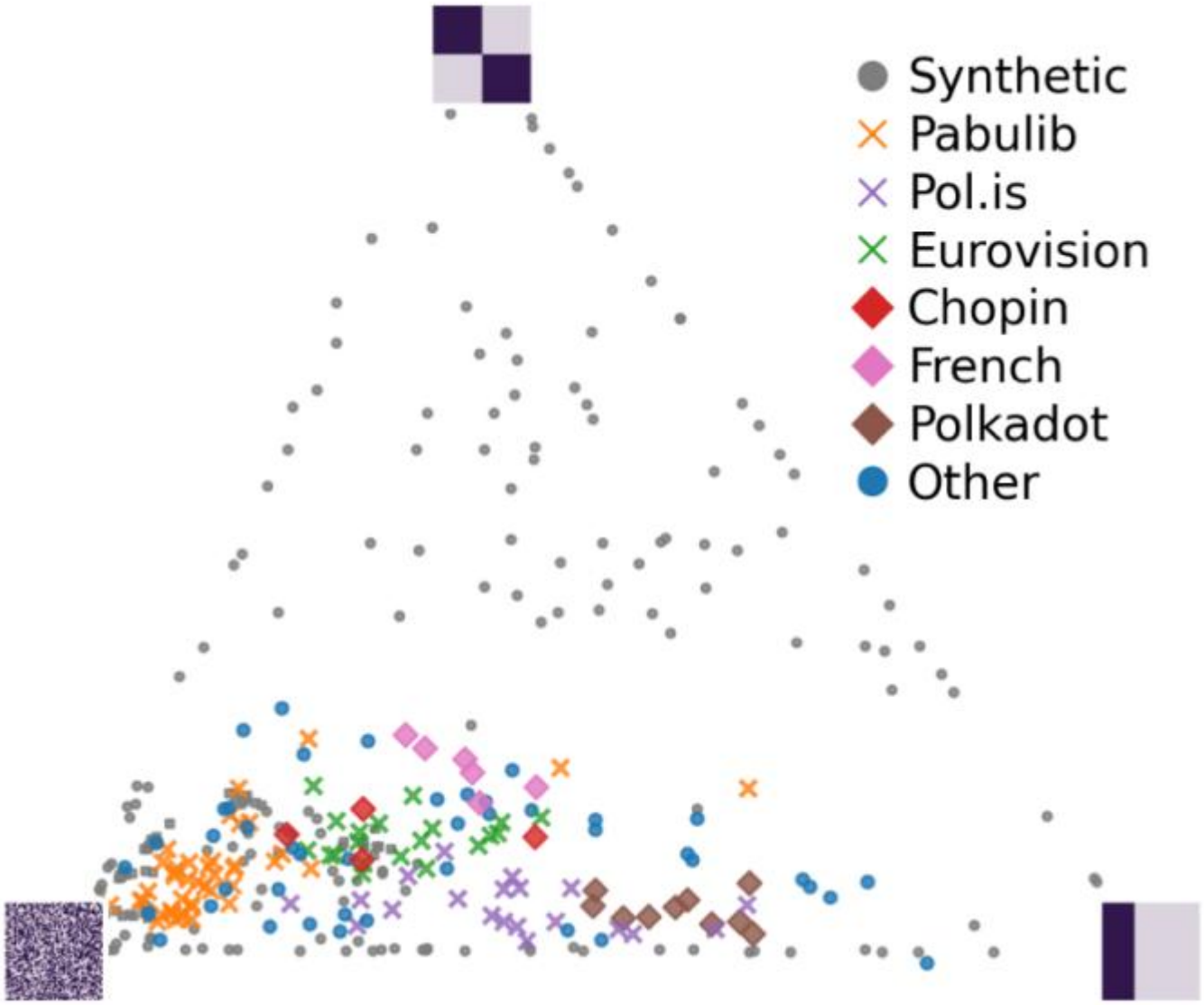
Approval Elections

Maps of Elections: Ordinal versus Approval

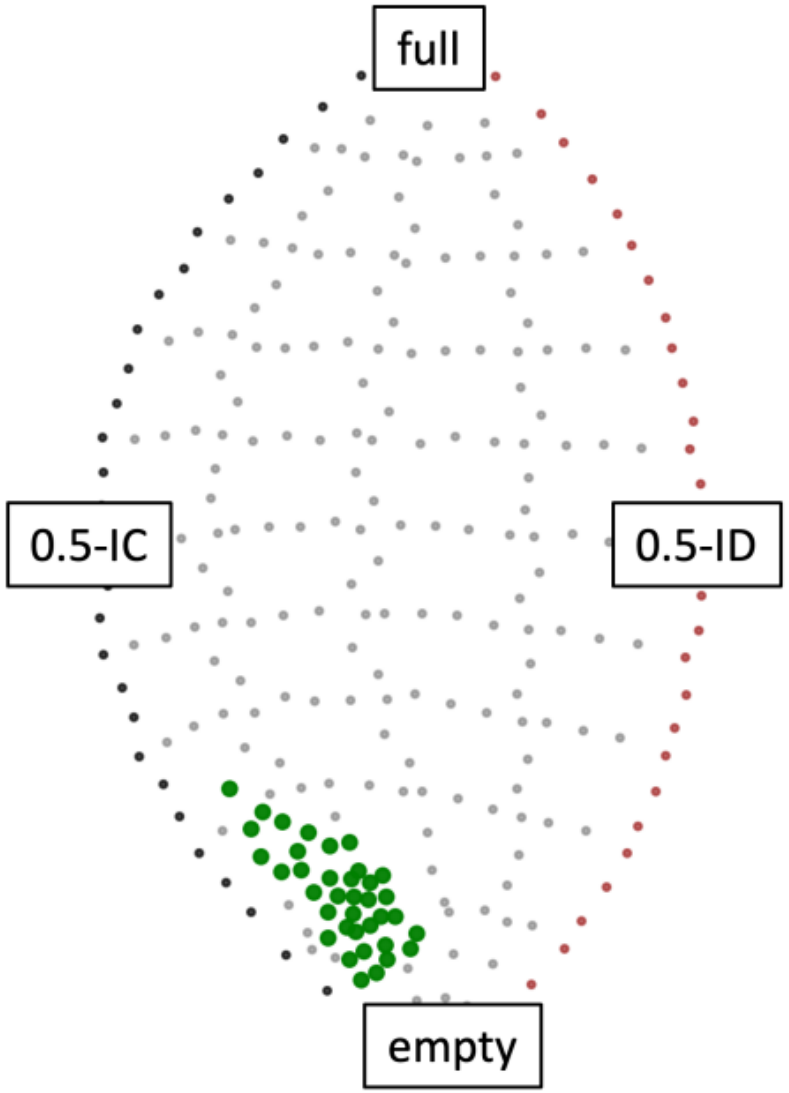


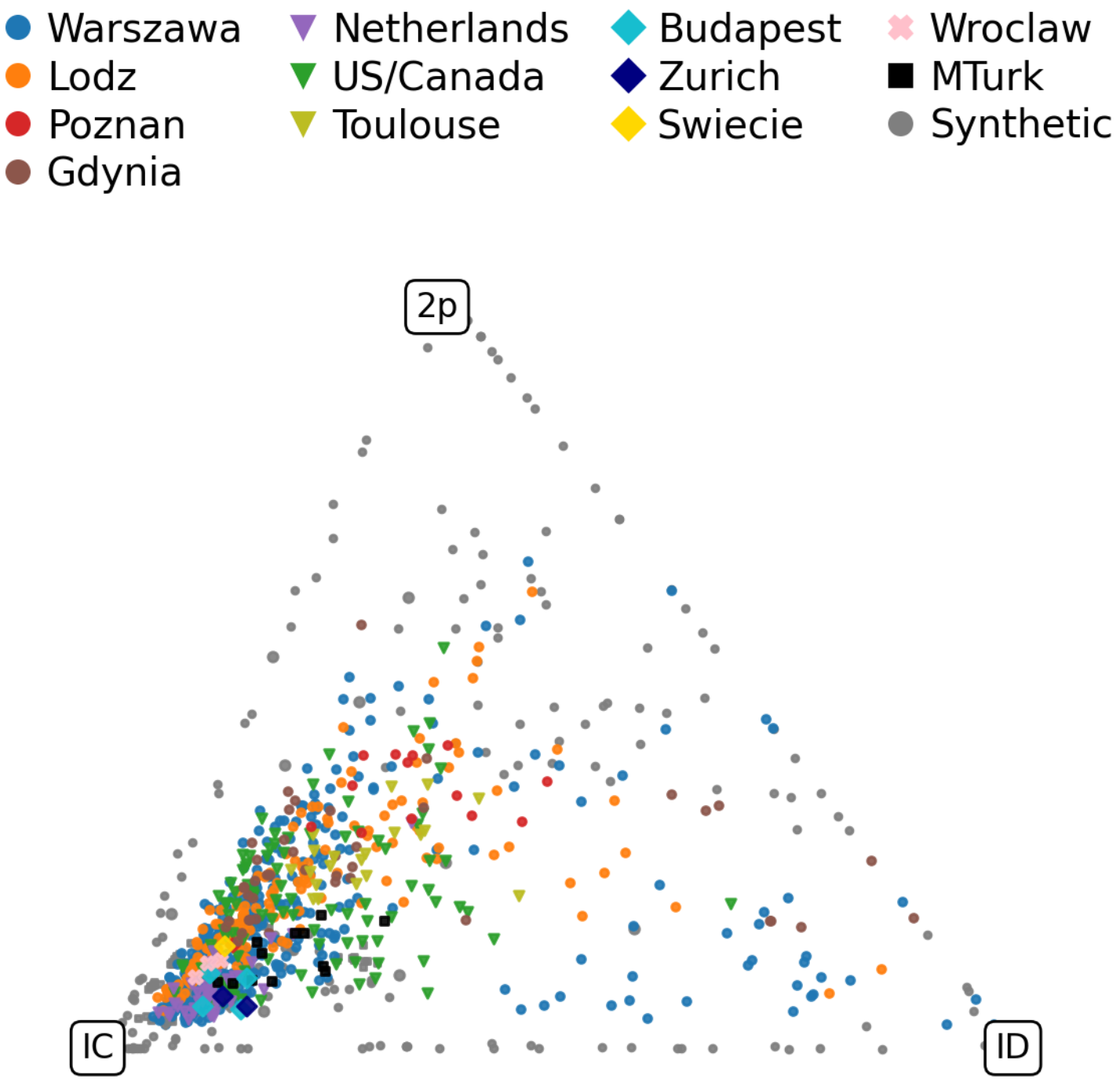
Map of Approval Elections



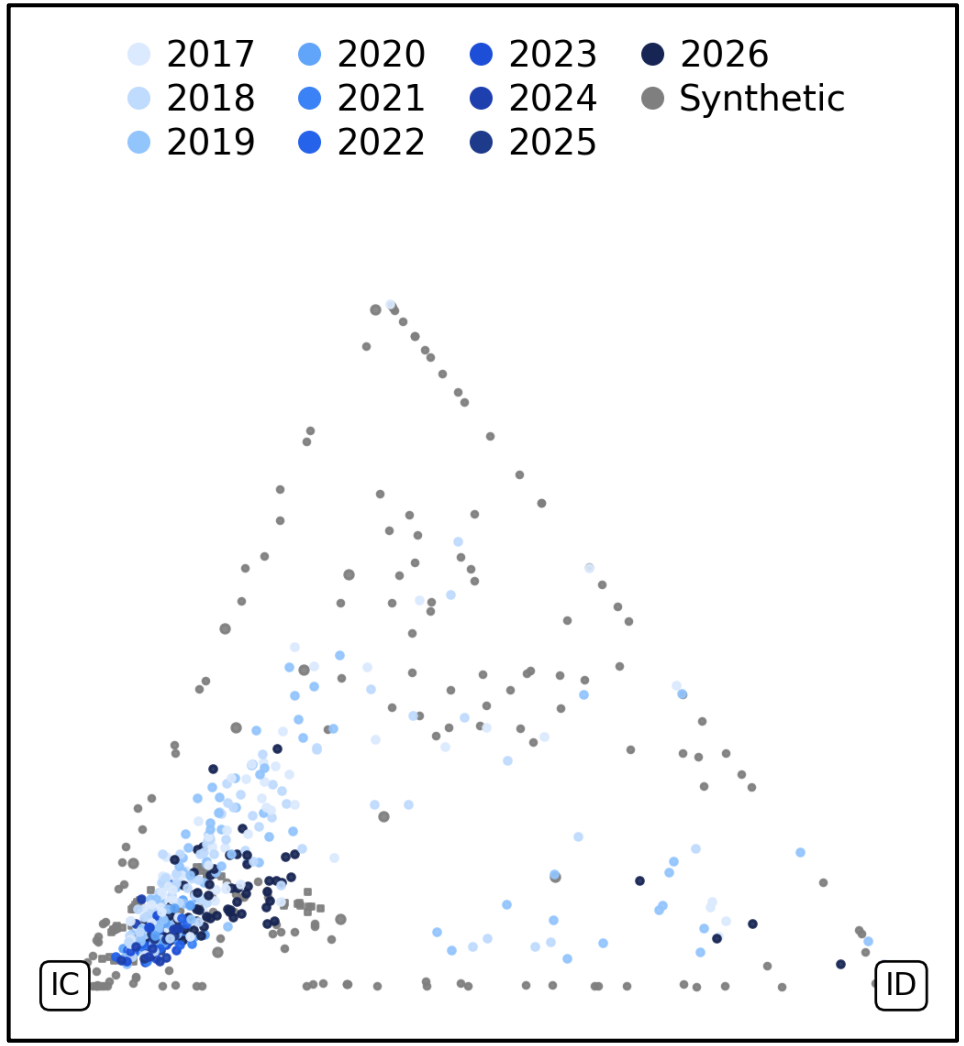


Map of Approval Elections

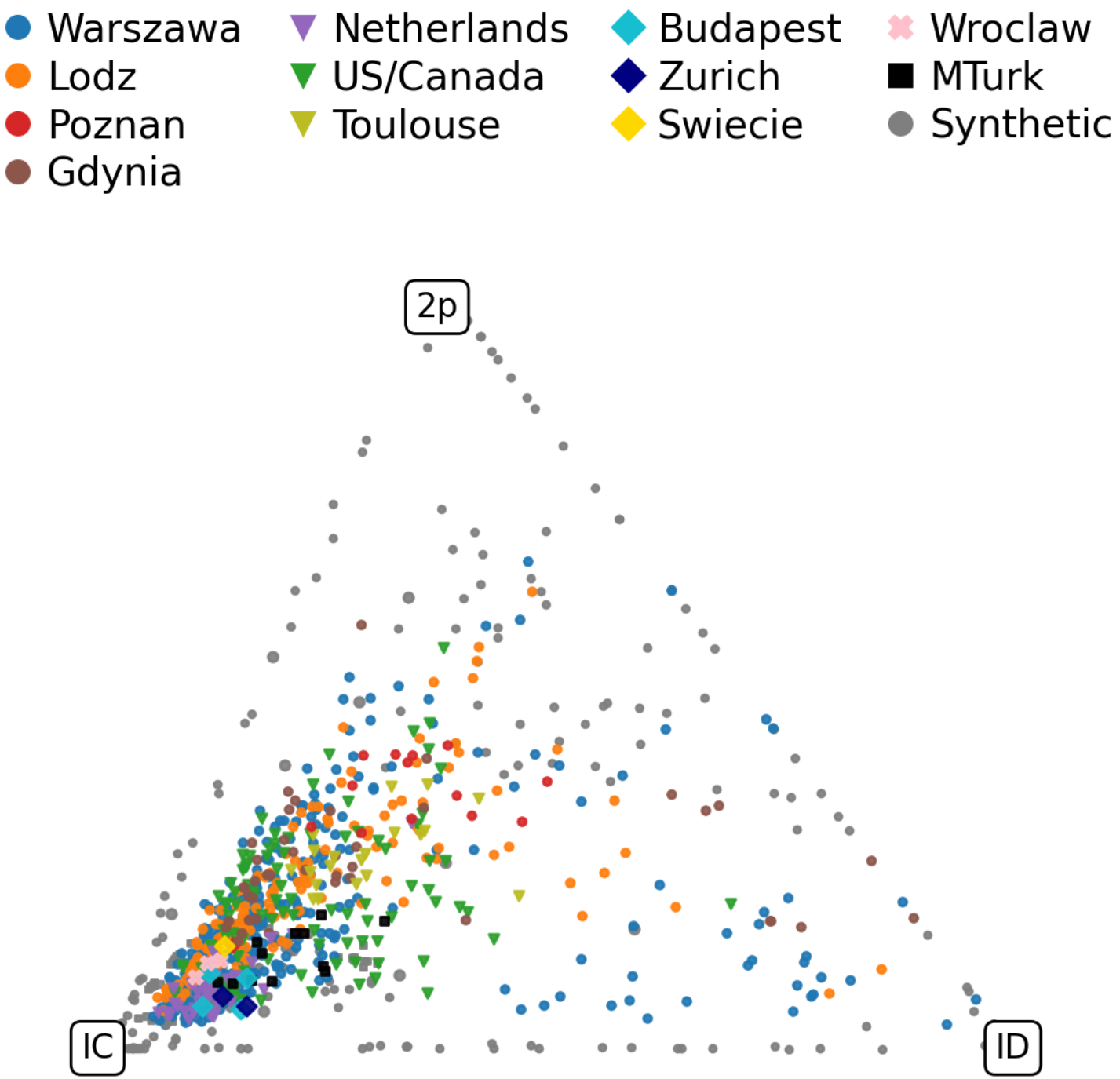




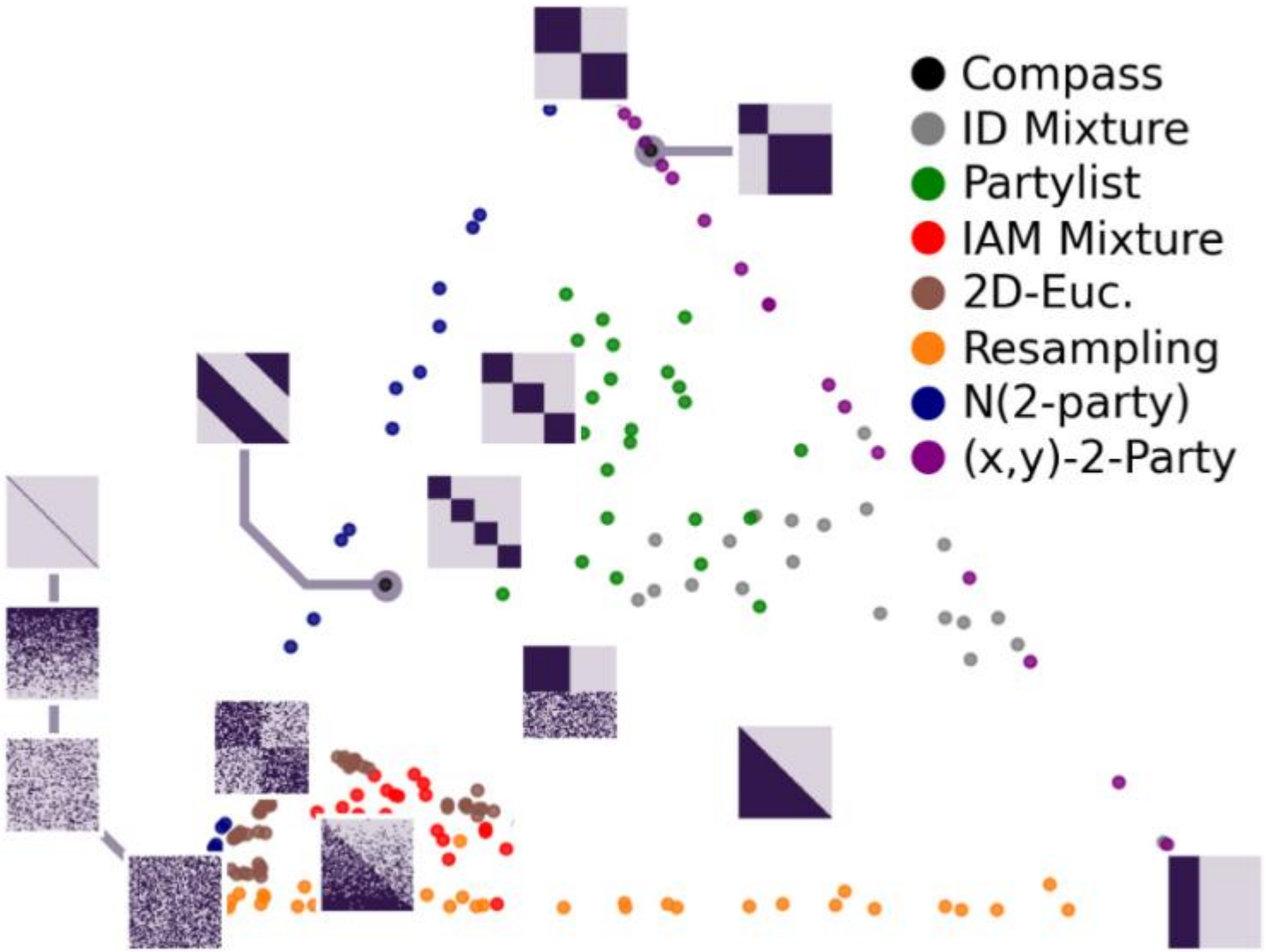
Map of Approval Elections

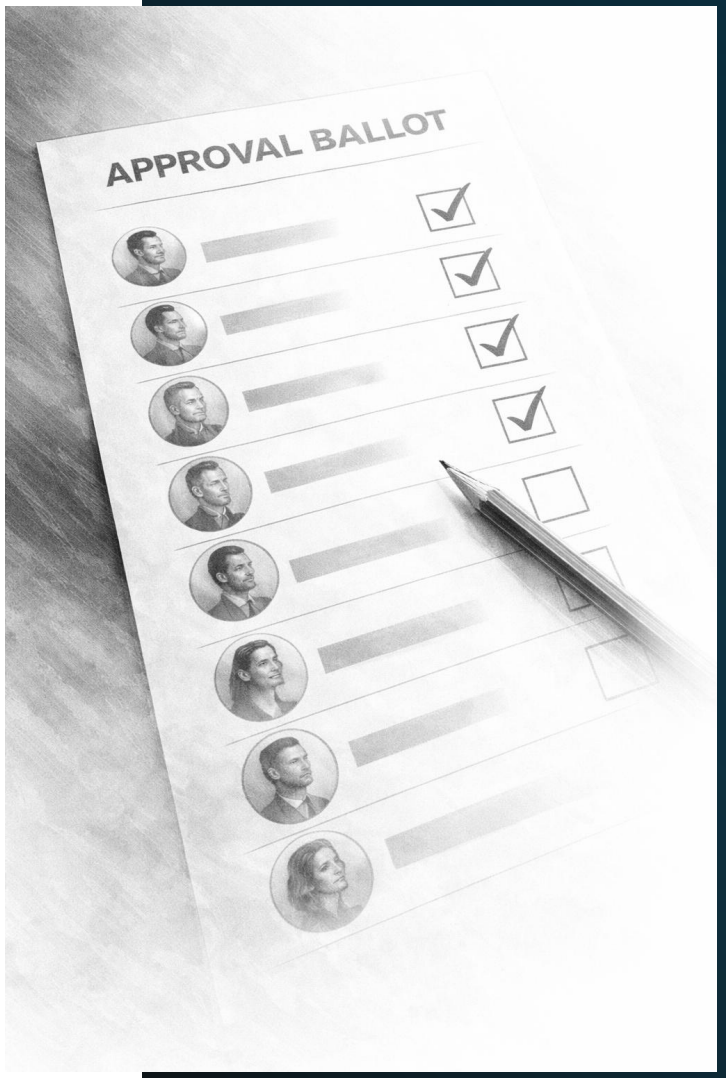


Warsaw PB



Map of Approval Elections



Conclusions

- Important features
  - Agreement
  - Diversity
  - Polarization
- Somewhat involved normalization
- Generate interesting/useful maps